

Boron Ligands Boosting the Electrochemiluminescence Performance of Europium Metal–Organic Frameworks by Facilitating the Electronic Bridging

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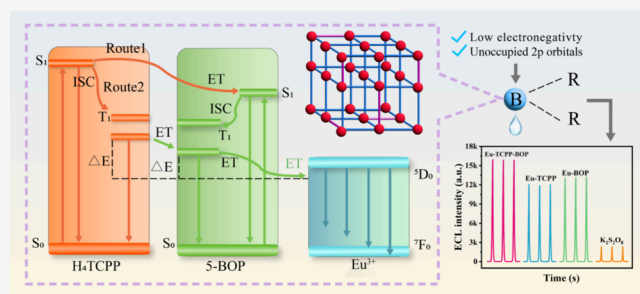


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Supporting Information

ABSTRACT: For optimal energy transfer in self-luminous lanthanide metal–organic frameworks (Ln-MOFs), the energy of the lowest triplet excited state must align with ideal energy levels. Failure to meet this condition can lead to reverse energy transfer, reducing luminous efficiency. In this study, we developed a mixed-ligand MOF, Eu-TCPP-BOP, which exists as an ECL self-enhancing luminophore. We used SPECM to study the role of boron ligands as a bridge for electron transport in improving the ECL performance of Eu-TCPP. The ligands H₄TCPP and 5-BOP act as electron donor and shuttle, facilitating electron transport during the synthesis of Eu-TCPP-BOP and promoting energy transfer to the excited state of the acceptor Ln³⁺, thus enhancing overall energy transfer in Ln-MOF. The results indicate that the introduction of boron ligands enhances the ECL intensity of Eu-TCPP by a factor of 1.4 under voltage excitation. As an ECL sensing platform, it demonstrates high sensitivity and selectivity for the detection of catechol, with a concentration range of 1–70 μ M and a detection limit of 0.35 μ M.



Electrochemiluminescence (ECL) is a process in which free radicals generated on the electrode surface participate in an electron transfer reaction, resulting in the formation of an excited state that subsequently emits light.^{1–3} Its exceptional sensitivity, precise controllability, broad linear range, low background interference, and rapid response time have drawn significant interest across various fields, including clinical diagnostics, environmental monitoring, and food safety.^{4–6} Traditional ECL emitters can be classified into three main categories: metal complexes (such as Ru, Pt, and Os),^{7,8} aromatic organic compounds (like anthracene, porphyrin, perylene, and phenothiazine derivatives),^{9–13} and nanomaterials (including quantum dots and metal nanoclusters).¹⁴ Recently, highly crystalline metal–organic frameworks (MOFs) composed of metal ions or clusters and organic ligands linked by coordination bonds have gained recognition as ECL emitters.¹⁵ These MOFs enhance the reaction rates between emitters and coreactants or reduce the diffusion distance between them, thus improving the overall ECL efficiency.^{16–18} Self-luminous lanthanide metal–organic frameworks (Ln-MOFs) are particularly notable for their tunable luminescent properties, large Stokes shifts, extended decay times, stable emission energies, and robust energy transfer capabilities,^{19–21} making them highly suitable for ECL applications.^{22–24} However, if the energy levels are not ideally

aligned, reverse energy transfer may occur, diminishing their luminescence efficiency.

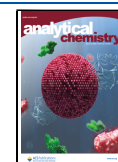
The low electronegativity and the presence of unoccupied 2p orbitals of boron make it a unique electron acceptor with notable electron affinity and strong Lewis acidity.^{25,26} When boron is incorporated into an electron-rich π -conjugated system, electron delocalization into the vacant 2p orbitals of boron reduces the exchange of electrons between donor and acceptor components.^{27,28} To address this limitation, Chen et al. developed a tetrad-boron system with two electron donors and one acceptor, which exploits the interaction between the acceptor's lone electron pair and boron's vacant p orbital. This design enhances the electron-accepting capacity of the acceptor while stabilizing the electron configuration of boron.^{29,30} In this context, the electron-rich boron atoms facilitate electron transfer to donor components, thus boosting their electron-donating ability.^{31,32} Incorporating boron-containing ligands into Ln-MOFs helps achieve the ideal

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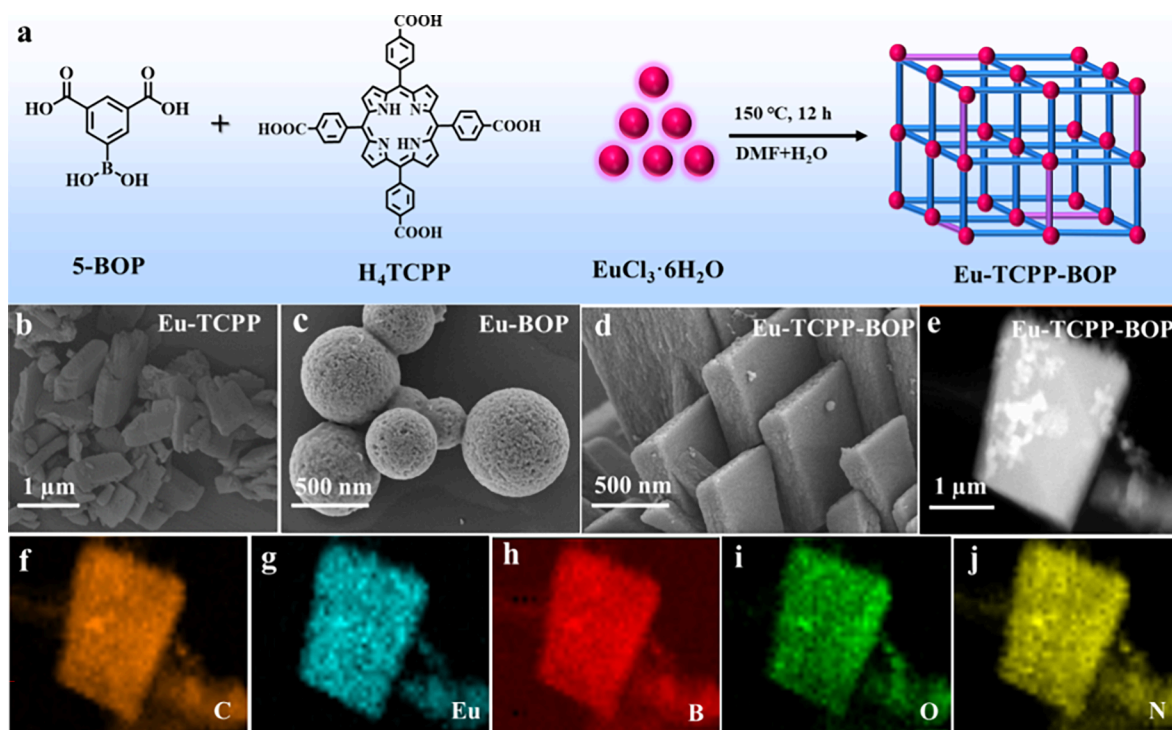


Figure 1. (a) Schematic diagram of the synthesis of Eu-TCPP-BOP; the SEM images: (b) Eu-TCPP, (c) Eu-BOP, (d) Eu-TCPP-BOP; TEM images: (e) Eu-TCPP-BOP; (f–j) elemental mapping of Eu-TCPP-BOP.

lowest triplet excited state energy, optimizing energy levels for maximum energy transfer.³³ Boron acts as a crucial electron shuttle, fine-tuning the energy transfer pathways between lanthanide ions (Ln^{3+}) and ligands, enhancing donor–acceptor interactions, and narrowing the energy band gap. This narrowing is essential for improving energy transfer efficiency within the Ln-MOF framework.³⁴

This study introduces the mixed-ligand MOF, Eu-TCPP-BOP, which enhances autoluminescence through electrochemiluminescence. We employed the aggregation-induced emission (AIE) ligand, Meso-Tetra (4-carboxyphenyl) porphyrine (H_4TCPP), alongside 5-boronoisophthalic acid (5-BOP) as the primary and secondary ligands, respectively. Scanning photoelectrochemical microscopy (SPECM) was employed to investigate the role of boron ligands as a bridge for electron transmission in enhancing the ECL performance of Eu-TCPP. In this setup, both H_4TCPP and 5-BOP act as intermediaries for electron donation and transport during the synthesis of Eu-TCPP-BOP, facilitating electron energy transfer to the excited state of the acceptor (Ln^{3+}), thereby optimizing energy transfer within the Ln-MOF. The empty 2p orbital of boron imparts significant Lewis acidity to boron-containing compounds. In the presence of catechol, the sensitization of the ligands (H_4TCPP and 5-BOP) to Eu^{3+} is disrupted, leading to a reduction in the ECL signal. This phenomenon establishes an ECL system characterized by a low detection limit and high selectivity for catechol monitoring.

EXPERIMENTAL SECTION

Reagents and Materials. Europium(III) chloride hexahydrate ($\text{EuCl}_3 \cdot 6\text{H}_2\text{O}$), potassium persulfate ($\text{K}_2\text{S}_2\text{O}_8$), potassium chloride (KCl), potassium dihydrogen phosphate (KH_2PO_4), dipotassium hydrogen phosphate (K_2HPO_4), 4-formylbenzoic acid ($\text{C}_8\text{H}_6\text{O}_3$), potassium hexacyanoferrate

$[\text{K}_4\text{Fe}(\text{CN})_6]$ and potassium ferricyanide $[\text{K}_3\text{Fe}(\text{CN})_6]$ were purchased from Aladdin Biochemical Technology Co., Ltd. (Shanghai, China). 5-boronoisophthalic acid ($\text{C}_8\text{H}_7\text{BO}_6$, 5-BOP), phenylboronic acid ($\text{C}_6\text{H}_7\text{BO}_2$, PBA), 4-carboxyphenylboronic acid ($\text{C}_7\text{H}_7\text{BO}_4$, CPBA), 4-aminophenylboronic acid ($\text{C}_6\text{H}_8\text{BNO}_2$, APBA) and catechol ($\text{C}_6\text{H}_6\text{O}_2$) were purchased from Maclin Biochemical Technology Co., Ltd. (Shanghai, China). N, N-dimethylformamide (DMF), propionic acid ($\text{C}_3\text{H}_6\text{O}_2$), and pyrrole ($\text{C}_4\text{H}_5\text{N}$) were purchased from Zesheng technology Co., Ltd. (Anhui, China).

Apparatus. Material characterization equipment mainly includes the following instruments: AV 400 MHz nuclear magnetic resonance spectrometer for recording ^1H -NMR spectrum (^1H -NMR, Bruker Instrument Co., Ltd., Switzerland), Tecnai G2 F20 transmission electron microscope (TEM, FEI Company, USA), Quanta 200 scanning electron microscope (SEM, Philips FEI Company, Netherlands), Digilab FTS-3000 Fourier transform infrared spectrum (FT-IR, Nicolet, USA), Bruker D8 ADVANCE powder X-ray diffraction (XRD, Bruker Co., Ltd., Germany), and PHI5000 VersaProbe III X-ray photoelectron spectroscopy (XPS, UIVAC-PHI, Japan). The experimental testing equipment mainly includes MPI-A capillary electrophoresis electrochemiluminescence analyzer (Remax Analytical Instruments Co., Ltd., Xi'an) with a photomultiplier tube bias of 800 V, electrochemiluminescence spectrometer (ECLS-ML, FORTEC Technology (HK) Co. Ltd., Hong Kong, China), UV-102 ultraviolet spectrophotometer (UV-vis, Tianmei Scientific Instruments Co., Ltd., Shanghai), and LS-55 fluorescence spectrophotometer (PL, PerkinElmer, USA). The electrode characterization device is a CHI 600e electrochemical workstation (Chenhua Instrument Co., Ltd., Shanghai).

Preparation of Eu-TCPP-BOP. According to the literature,²² $\text{EuCl}_3 \cdot 6\text{H}_2\text{O}$ (0.1466 g), H_4TCPP (0.1005 g, Figure S1) and 5-BOP (0.021, 0.042, 0.063, 0.084, 0.105 and 0.126 g)

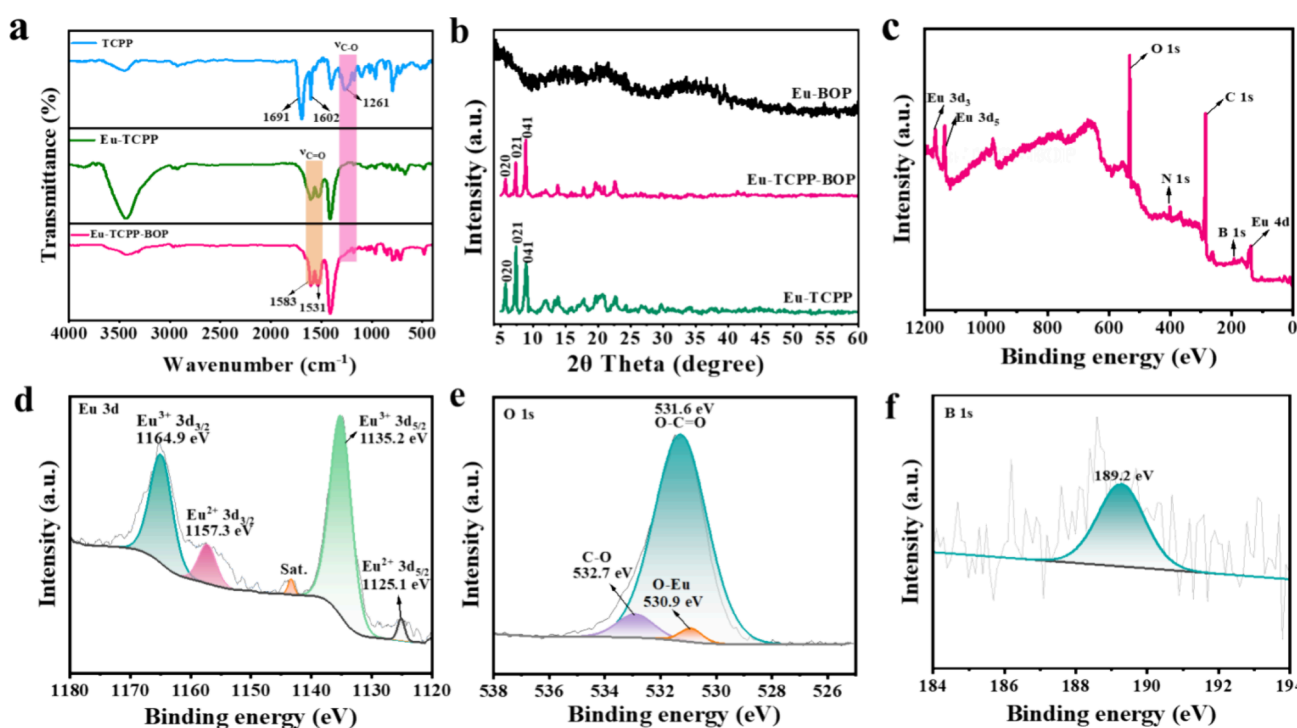


Figure 2. (a) FT-IR: TCPP, Eu-TCPP, and Eu-TCPP-BOP; (b) XRD: Eu-BOP, Eu-TCPP-BOP, and Eu-TCPP; (c) XPS full spectrum of Eu-TCPP-BOP; (d–f) high-resolution spectra of Eu 3d, O 1s, and B 1s.

were mixed with 40 mL of DMF/H₂O (7:3, V/V) and ultrasound for 15 min. The mixture was poured into a Teflon-lined stainless-steel autoclave and heated in an oven at 150 °C for 12 h. The mixture was cooled to room temperature, washed with DMF and ethanol at least 3 times to remove excess ligands and metal ions. Finally, the samples were dried in an oven at 70 °C for 12 h to obtain the black-brown Eu-TCPP-BOP powder.

Electrochemical and ECL Experiments. All electrochemical experiments were performed using a three-electrode system, including glass carbon electrode (GCE) with a diameter of 3 mm as the working electrode. Platinum wire was used as a counter electrode and an Ag/AgCl reference electrode (containing saturated KCl solution) served as the reference electrode. A buffer solution (0.1 M, pH = 7.4) containing K₂HPO₄, KCl, and KH₂PO₄ was used at room temperature. The MPI-A ECL instrument was used to preliminarily evaluate the ECL performance of Eu-TCPP-BOP. The related parameters of the ECL test are as follows: the voltage of the photomultiplier tube (PMT) of 800 V, the scanning speed of 0.2 V/s, and the potential window of 0.0–1.8 V (vs Ag/AgCl). At room temperature, a buffer solution (0.1 M, pH = 7.4) containing K₂HPO₄, KCl, K₂S₂O₈, and KH₂PO₄ was used for testing. For ECL spectra, it is essential to cool the CCD camera to –60 °C prior to use. The spectrometer (ECLS-ML) and CCD camera settings were managed through the Andor Solis program, and wavelength calibration was conducted using commercial mercury lamps. Following this, the luminophore (0.004 mg/mL) was added to a pH = 7.4, 0.1 M phosphate buffer solution containing K₂S₂O₈. The electrochemical cell containing this solution was then placed in the spectrometer, and the CCD camera was connected to obtain phase three-dimensional and two-dimensional ECL spectral data. For SPECM measurement, the electrodes of platinum (Pt) and silver/silver chloride (Ag/

AgCl) were used as counter electrode and reference electrode, respectively. Ultramicroelectrode (UME) with a diameter of 25 μm and FTO electrodes modified with Eu-TCPP, Eu-BOP and Eu-TCPP-BOP were used as working electrodes. Throughout the measurement process, K₃Fe(CN)₆ was used as a probe molecule, and its electrochemical performance was studied in a 0.1 M KCl solution. The feedback mode was used for dynamic studies, and the rate constant (*K*_{eff}) of the working electrode was determined.

RESULTS AND DISCUSSION

Characterization of Eu-TCPP-BOP. Figure 1a illustrates the synthesis process of Eu-TCPP-BOP using a solvothermal method.²² The morphologies of Eu-BOP, Eu-TCPP, and Eu-TCPP-BOP were examined via scanning electron microscopy (SEM) and transmission electron microscopy (TEM). As shown in Figure 1b–d, the three compounds exhibit distinct shapes: Eu-BOP forms spherical structures, Eu-TCPP appears as irregular blocks, and Eu-TCPP-BOP adopts a sheet-like configuration, each with varying dimensions. Notably, Figure 1e reveals a transformation in the shape of Eu-TCPP from block-like to a more rectangular structure upon the introduction of the secondary ligand, 5-BOP. The elemental mapping presented in Figures 1f–j confirms the successful integration of 5-BOP into the Eu-TCPP framework, clearly displaying the presence of C, Eu, B, O, and N.

Figure 2 details the use of powder X-ray diffraction (XRD), X-ray photoelectron spectroscopy (XPS), and Fourier transform infrared spectroscopy (FT-IR) to investigate the crystal structure, bonding configuration, and characteristic absorption bands associated with functional groups in Eu-TCPP-BOP. In the FT-IR spectrum (Figure 2a), the asymmetric stretching vibration of the –C=O group in H₄TCPP shifts from 1691 and 1602 cm^{–1} to 1583 and 1531 cm^{–1}, respectively. Additionally, the absorption peak for –C=O at 1695 cm^{–1},

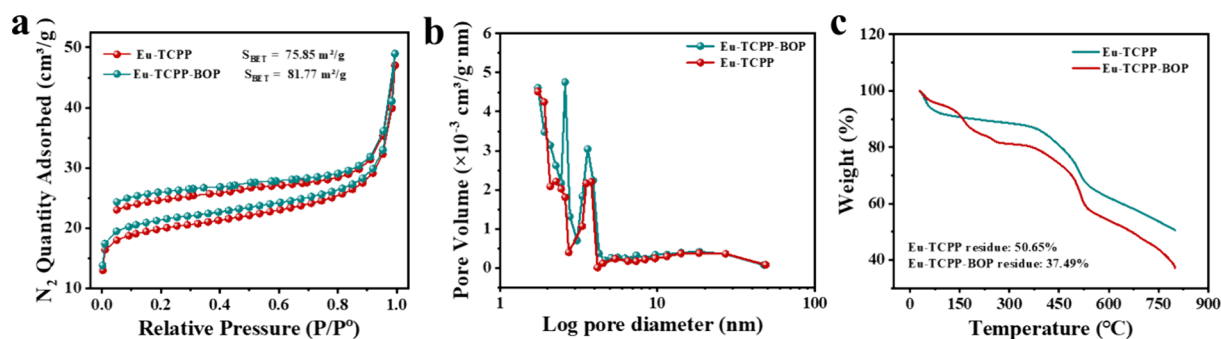


Figure 3. (a, b) N₂ adsorption–desorption isotherms of Eu-TCPP-BOP and corresponding pore-size distribution curves; (c) TGA curves of Eu-TCPP-BOP measured in a N₂ atmosphere at a temperature ramp of 10 °C/min.

seen in the ligand, disappears in Eu-BOP (Figure S2a),³⁵ while the stretching vibration of –C–O in TCPP and 5-BOP at 1261 cm⁻¹ is no longer present, suggesting effective coordination of –COOH with Eu³⁺.³⁶ The XRD pattern (Figure 2b) confirms the crystal structure of Eu-TCPP-BOP, showing 2θ values and crystal planes corresponding to diffraction peaks at 5.79° (020), 7.32° (021), and 8.83° (041).³⁷ Furthermore, the crystal framework of Eu-TCPP remains stable with the incorporation of 5-BOP, indicating that the auxiliary ligand primarily influences the morphology rather than the overall crystal structure of Eu-TCPP.

The full X-ray photoelectron spectroscopy (XPS) spectrum of Eu-TCPP-BOP confirms the presence of Eu, N, C, O, and B elements (Figure 2c). In the high-resolution Eu 3d spectrum, a peak at 1143.4 eV is identified as a satellite signal (“Sat”), while the peaks at 1164.9 and 1135.2 eV correspond to Eu³⁺ (Figure 2d).³⁸ A small amount of Eu²⁺ is also detected at 1157.3 and 1125.1 eV, attributed to a surface valence transition. The C 1s high-resolution spectrum (Figure S2b) shows three distinct peaks at 288.4, 285.9, and 284.8 eV, representing the –O–C=O, –C–N, and –C–C bonds, respectively.³⁹ The high-resolution N 1s spectrum (Figure S2c) reveals binding energies of 409.1 eV for the –N–H bond and 399.9 eV for the –C–N bond.³⁸ These high-resolution N 1s and C 1s analyses confirm that the electron-deficient boron group does not interact with nitrogen in H₄TCPP. In the O 1s high-resolution spectrum (Figure 2e), three distinct peaks are observed: the peak at 532.7 eV is attributed to the –C–O bond, 531.6 eV corresponds to the –COOH bond, and 530.9 eV is associated with the –Eu–O bond, indicating effective coordination between –COOH and Eu³⁺.⁴⁰ The binding energy of B 1s (Figure 2f) at 182.9 eV further supports the successful incorporation of the 5-BOP ligand into the Eu-TCPP structure.

With the integration of 5-BOP, the BET (Brunauer–Emmett–Teller) surface area increased from 75.85 m²/g to 81.77 m²/g (Figure 3a). The pore distribution graph (Figure 3b) indicates that the pore size in Eu-TCPP-BOP remains comparable to that of Eu-TCPP. Thermogravimetric analysis (TGA) shows that both MOFs exhibit a weight loss of approximately 20% between 25 and 350 °C, which is attributed to the evaporation of coordinated solvent molecules. A significant weight reduction at around 350 °C corresponds to the decomposition of the H₄TCPP and 5-BOP ligands, indicating that the structural integrity of the material begins to degrade at this temperature (Figure 3c).

Photophysical Properties of H₄TCPP, Eu-TCPP, Eu-BOP, and Eu-TCPP-BOP. In Figure S3a, the absorption

spectrum of H₄TCPP is presented, showing a prominent peak at 423 nm corresponding to the Soret band, which is associated with the a_{1u}(π)–e_g^{*}(π) transition. Additionally, peaks at 515, 549, 590, and 646 nm are observed, representing the four Q bands resulting from the a_{2u}(π)–e_g^{*}(π) transition. The fluorescence spectra for Eu-TCPP, Eu-BOP, and Eu-TCPP-BOP are displayed in Figures S3b,c. Upon excitation at 300 nm, a Eu²⁺ transition from 4f⁷ to 4f⁶5d¹ is observed at 334 nm, along with the Eu³⁺ ⁵D₀ → ⁷F_J transitions (where J = 1, 2) at 600 and 662 nm.⁴¹

To further examine the photophysical behavior of these materials, the emission decay curve of Eu³⁺ at 653 nm (excited at 340 nm) was analyzed. The duration of the excited state of a fluorescent substance determines the value of its fluorescence lifetime. This lifetime can be viewed as a state function, remaining unaffected by various factors such as the wavelength of excitation, the duration of light exposure, whether the excitation is single-photon or multiphoton, and it is also not impacted by photobleaching.⁴² Additionally, the fluorescence lifetime serves as a metric that does not correlate with the intensity of luminescence or the concentration levels of fluorophores. Given its association with energy unstable states, this lifetime is influenced by various internal and external factors related to the structure of the fluorophore, including temperature, polarity, and the presence of fluorescence quenchers.⁴³ The fluorescence lifetimes were calculated using eq 1, where I₀ is the initial intensity. *t* is the time variable, and τ represents the decay time.⁴⁴

$$I = I_0 \exp\left(-\frac{t}{\tau}\right) \quad (1)$$

The lifetimes of the 4f → 4f transitions for Eu-TCPP, Eu-BOP, and Eu-TCPP-BOP at 653 nm are recorded as 0.26 ms, 6.9 μs, and 8.4 μs, respectively (Figure S3d–f). Notably, the incorporation of the auxiliary ligand 5-BOP increases the lifetime of Eu-TCPP from 6.9 to 8.4 μs, indicating that 5-BOP enhances ligand sensitization to Eu³⁺.

Electrochemical and ECL Performance of Eu-TCPP, Eu-BOP, and Eu-TCPP-BOP. ECL coreactants produce highly oxidizing or reducing intermediates through electron transfer in redox reactions. These intermediates interact with either oxidized or reduced ECL luminescent species, forming excited luminescent states, which return to the ground state and emit light. The incorporation of 5-BOP into Eu-TCPP makes Eu-TCPP-BOP act as a Lewis acid, significantly enhancing its solubility in aqueous environments. Contact angle measurements reveal values of 61° for Eu-TCPP and 32.8° for Eu-TCPP-BOP (Figure 4). The reduced contact

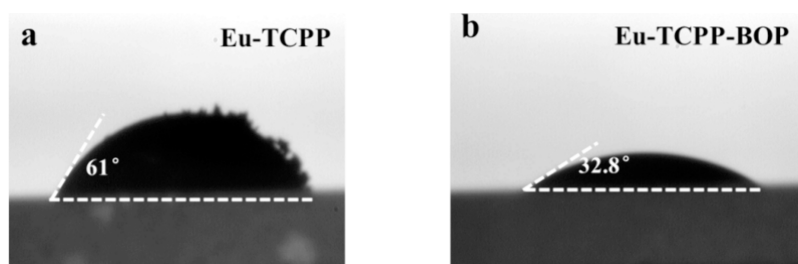


Figure 4. (a, b) Contact angle of Eu-TCPP and Eu-TCPP-BOP.

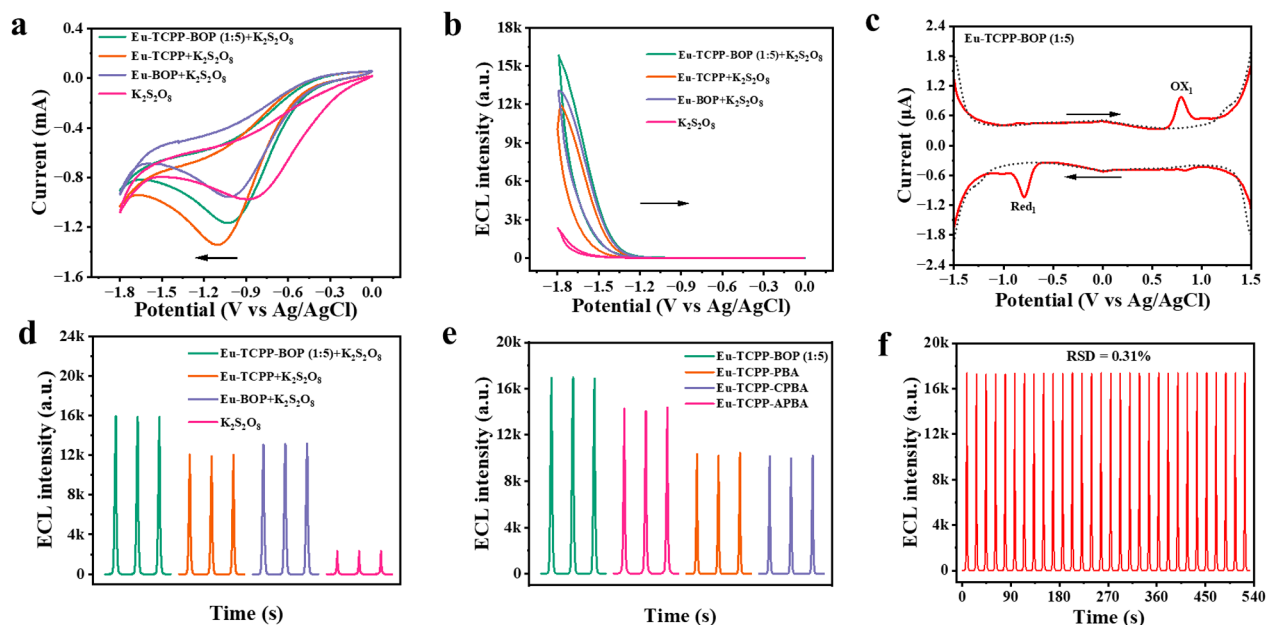


Figure 5. (a, b) Cyclic voltammetry and ECL-potential curve of Eu-TCPP, Eu-BOP and Eu-TCPP-BOP; (c) DPV curves of Eu-TCPP-BOP; (d) ECL-time curve of Eu-TCPP, Eu-BOP, and Eu-TCPP-BOP; (e) ECL-time curve of Eu-TCPP-PBA, Eu-TCPP-APBA, Eu-TCPP-CPBA, and Eu-TCPP-BOP; (f) ECL-time curve after 30 cycles of Eu-TCPP-BOP. Experimental conditions: 0.1 M $K_2S_2O_8$, 0.1 M PBS/KCl (pH = 7.4), scanning rate 0.2 V/s, potential window 0.0 ~ -1.8 V (vs Ag/AgCl), the concentration of Eu-TCPP, Eu-BOP, and Eu-TCPP-BOP was 0.004 mg/mL.

angle of Eu-TCPP-BOP reflects its superior surface hydrophilicity, facilitating better interactions with the aqueous medium and significantly enhancing ECL efficiency.

To further investigate the ECL performance of Eu-TCPP-BOP in aqueous conditions, the effects of various coreactants, including $K_2S_2O_8$, hydrogen peroxide, tripropylamine, and triethylamine, were evaluated. As shown in Figure S4a, $K_2S_2O_8$ notably enhances the ECL signal for Eu-TCPP-BOP, while other coreactants do not show substantial improvement. To optimize the ECL performance of Eu-TCPP-BOP in aqueous media, experimental parameters such as $K_2S_2O_8$ concentration, pH, potential window, scanning speed, and the H_4TCPP :5-BOP ratio were fine-tuned. With $K_2S_2O_8$ as the coreactant (Figure S4b–f), the optimal conditions were determined to be 0.1 M $K_2S_2O_8$, a pH of 7.4, a potential range from 0.0 ~ -1.8 V (vs Ag/AgCl), a scan rate of 0.2 V/s, and a H_4TCPP :5-BOP ratio of 1:5.

A 5 mL phosphate buffer solution containing 0.1 M KCl and $K_2S_2O_8$ (pH = 7.4) was introduced into the electrolytic cell. Subsequently, DMF-dissolved Eu-TCPP (0.004 mg/mL), Eu-BOP (0.004 mg/mL), and Eu-TCPP-BOP (0.004 mg/mL) were added to investigate their ECL properties. As illustrated in Figure 5, the ECL performance of Eu-TCPP, Eu-BOP, and Eu-TCPP-BOP was assessed using CV and ECL potential curves. It is generally observed that the closer the redox

potential of the coreactant is to that of the luminescent species, the more rapid the electron transfer occurs, resulting in a stronger ECL signal. Our experiments determined the reduction potential of Eu-TCPP to be -1.1 V (vs Ag/AgCl) (Figure 5a). Upon the introduction of 5-BOP into Eu-TCPP, it was found that the reduction potential of the Eu-TCPP-BOP complex was -1.0 V (vs Ag/AgCl), which is closer to the reduction potential of $K_2S_2O_8$ (-0.9 V vs Ag/AgCl). Consequently, the electron transfer in the catalytic reduction reaction of Eu-TCPP-BOP is faster than that of Eu-TCPP (-1.1 V vs Ag/AgCl), thereby enhancing the ECL signal (Figure 5b).⁴⁵ The ultraviolet–visible diffuse reflectance spectra (UV–vis DRS) presented in Figure S5a–d for Eu-TCPP, Eu-BOP, and Eu-TCPP-BOP indicate their respective band gaps of 1.77, 3.22, and 1.63 eV. The incorporation of boron enhances the donor–acceptor interaction and effectively reduces the energy band gap. Compared to Eu-TCPP and Eu-BOP, the smaller band gap of Eu-TCPP-BOP facilitates electron transitions from the ground state to the excited state. This phenomenon also promotes radiative transitions as the excited state returns to the ground state, thereby enhancing luminescence.⁴⁶

The electrochemical behavior of Eu-TCPP (0.004 mg/mL), Eu-BOP (0.004 mg/mL), and Eu-TCPP-BOP (0.004 mg/mL) was investigated in a phosphate buffer solution containing 0.1

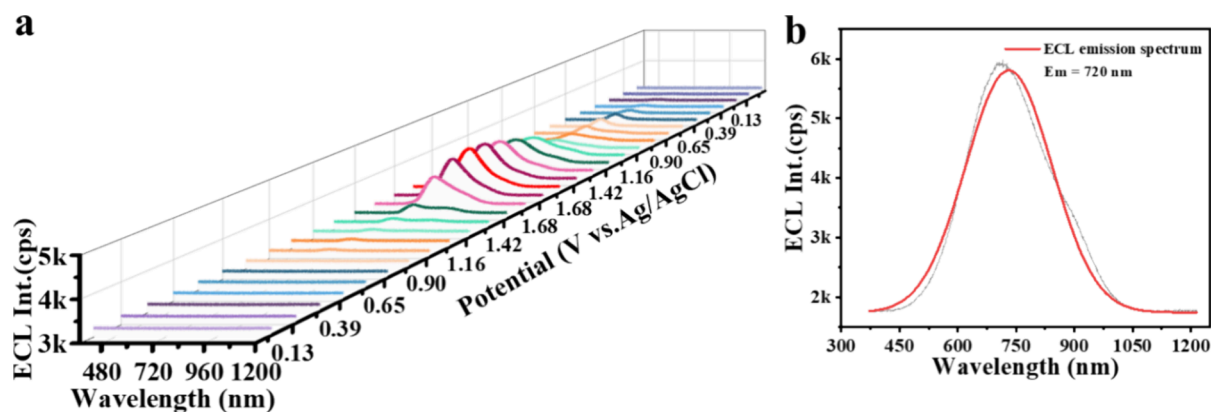


Figure 6. (a) Spooling ECL spectroscopy of Eu-TCPP-BOP; (b) ECL spectrum of 0.004 mg/mL Eu-TCPP-BOP. Experimental conditions: 0.1 M $K_2S_2O_8$, 0.1 M PBS/KCl (pH = 7.4), scanning rate 0.2 V/s, potential window 0.0 ~ -1.8 V (vs Ag/AgCl).

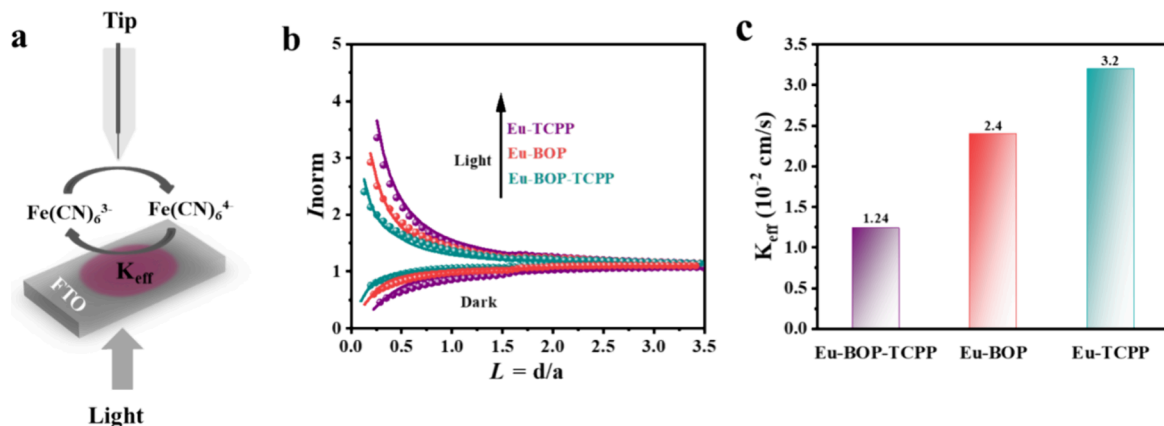


Figure 7. (a) Principle of the SPECM setup; (b) probe approach curves for the different modified substrates; (c) rate constants (K_{eff}) of different photoanodes.

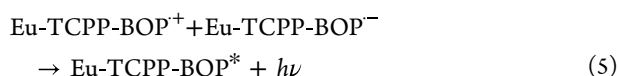
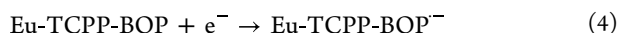
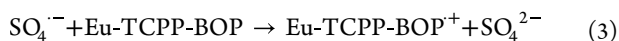
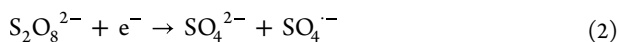
M KCl (pH = 7.4) using differential pulse voltammetry (DPV) under nitrogen atmosphere (Figure 5c and Figure S6a,b). The experimental setup was as follows: a scan rate of 0.2 V/s with a potential range of -1.5 ~ 1.5 V (vs Ag/AgCl). Unlike the bare GCE, which showed no notable redox peaks, Eu-TCPP, Eu-BOP, and Eu-TCPP-BOP exhibited pronounced redox peaks in the potential range of -1.5 ~ 1.5 V (vs Ag/AgCl). Reduction peaks were observed at -0.82 V for Eu-TCPP, -0.79 V for Eu-BOP, and -0.76 V for Eu-TCPP-BOP. Notably, Eu-BOP showed a more positive reduction potential than Eu-TCPP, suggesting an easier reduction process. Upon functionalization of Eu-TCPP with 5-BOP, the reduction peak shifted from -0.82 V to -0.76 V (vs Ag/AgCl), confirming the successful incorporation of 5-BOP and its positive impact on reducing ability. In terms of redox reversibility, Eu-TCPP, Eu-BOP, and Eu-TCPP-BOP exhibit both anode and cathode potentials, with consistent cathode and anode currents. This consistency indicates that the free radicals of Eu-TCPP, Eu-BOP, and Eu-TCPP-BOP possess high stability.⁴⁵

ECL Characteristics of Eu-TCPP Functionalized with Different Phenylboronic Acid Ligands. We further explored the ECL properties of Eu-TCPP functionalized with different phenylboronic acid ligands, specifically phenylboronic acid (PBA), 4-aminophenylboronic acid (APBA), and 4-carboxyphenylboronic acid (CPBA). As illustrated in Figure S7, the structure of Eu-TCPP remained largely unchanged when functionalized with PBA and CPBA. However, functionalization with APBA induced a morphological shift

from a block configuration to a stacked cuboid structure, highlighting the effective binding of APBA with the Eu^{3+} metal node. According to Figure 5e, the ECL intensity of Eu-TCPP with the three ligand modifications follows this order: Eu-TCPP-BOP > Eu-TCPP-APBA > Eu-TCPP-PBA $\approx$ Eu-TCPP-CPBA. Notably, the ECL intensity of Eu-TCPP-APBA (~14,000 au) surpasses that of Eu-TCPP (~12,000 au) (Figure 5d), indicating that boron acts as an electron transfer bridge, improving donor-acceptor interactions and reducing the band gap. Under optimal conditions, Figure 5f demonstrates that Eu-TCPP-BOP exhibits excellent stability after 30 scan cycles, with a relative standard deviation (RSD) of 0.31%.

Real-Time ECL Measurements and Mechanism. Real-time measurements of ECL intensity and wavelength can be conducted using spooling ECL spectroscopy. As illustrated in Figure 6a, a gradual negative shift in the applied voltage to -1.03 V (vs Ag/AgCl) initiates the reduction of the persulfate ion ($\text{S}_2\text{O}_8^{2-}$) to generate the sulfate anionic free radical ($\text{SO}_4^{\bullet-}$) (eq 2). Simultaneously, the Eu-TCPP-BOP species undergoes oxidation by $\text{SO}_4^{\bullet-}$, forming the strong reducing agent Eu-TCPP-BOP $^{\bullet+}$ (eq 3). As the potential continues to shift negatively, Eu-TCPP-BOP is further reduced to Eu-TCPP-BOP $^{\bullet-}$ (eq 4). During this process, the excited species Eu-TCPP-BOP $^{\bullet+}$ and Eu-TCPP-BOP $^{\bullet-}$ return to their ground state, producing the excited Eu-TCPP-BOP * species, which emits strong ECL (eq 5). As the potential decreases further, the concentrations of Eu-TCPP-BOP $^{\bullet+}$ and Eu-TCPP-BOP $^{\bullet-}$

near the electrode increase, which in turn raises the concentration of Eu-TCPP-BOP* and enhances the ECL emission. When the applied voltage reaches -1.8 V (vs Ag/AgCl), the ECL emission from Eu-TCPP-BOP* reaches its maximum intensity, with a peak wavelength of 720 nm (Figure 6b). This represents a redshift of 67 nm compared to the photoluminescence (PL) peak at 653 nm (Figure S3c), which is attributed to the relaxation of the excited Eu-TCPP-BOP* species to the surface state.⁴⁷ As the potential is scanned back in the reverse direction, the ECL emission gradually diminishes, likely due to the depletion and reversion of Eu-TCPP-BOP* and Eu-TCPP-BOP²⁺.



The effect of intramolecular charge transfer on the ECL performance of the lanthanide metal–organic framework following the introduction of boron ligands was evaluated using SPECM (Figure 7a–c). Figure S8 shows the characterization of ultramicroelectrode (UME). Under visible light irradiation, the SPECM probe current increases in various samples containing tin oxide, as photogenerated holes can be collected and participate in the oxidation process across different photoanode surfaces. Conversely, the probe current decreases under dark conditions.²¹ In this experiment, the combination of the probe approach curves (PAC) and the kinetic rate constant (K_{eff}) reveals that Eu-TCPP-BOP exhibits lower positive feedback current and K_{eff} compared to Eu-TCPP under light conditions. This observation indicates that a greater number of photogenerated carriers “escape” from Eu-TCPP to the electrode surface. Conversely, under dark conditions, Eu-TCPP-BOP displays a pronounced negative feedback current, suggesting that the introduction of the boron ligand (acting as a charge transfer bridge) accelerates the migration rate of Eu-TCPP carriers at the cathode. By integrating the SPECM and ECL potential curves (Figure 5b), we demonstrate that the incorporation of boron ligand molecules facilitates electron injection at the cathode potential and enhances energy transfer between Eu^{3+} and H_4TCPP , thereby promoting ECL emission.

Energy-Transfer Routes of Eu-TCPP-BOP. The 5-BOP ligand plays a significant role in enhancing the ability of Eu^{3+} and the primary ligand to absorb and transport energy. In the case of single-ligand Eu-TCPP, it has been well-documented^{33,48,49} that energy is transferred from the first excited singlet state (S_1) of TCPP to its first excited triplet state (T_1) via intersystem crossing, followed by energy transfer from the T_1 state of H_4TCPP to the emission level of Eu^{3+} . For the mixed-ligand system Eu-TCPP-BOP, the energy transfer occurs through two primary pathways (Figure 8): (1) Energy is transferred from the S_1 state of TCPP to the S_1 state of 5-BOP, subsequently moving to the T_1 state of 5-BOP and finally reaching the $^5\text{D}_0$ level of Eu^{3+} ; (2) Energy is transferred from the S_1 state of H_4TCPP to its T_1 state via intersystem crossing, then moves from the T_1 state of H_4TCPP to the excited state of 5-BOP, and finally to the $^5\text{D}_0$ level of Eu^{3+} through the T_1 state of 5-BOP.

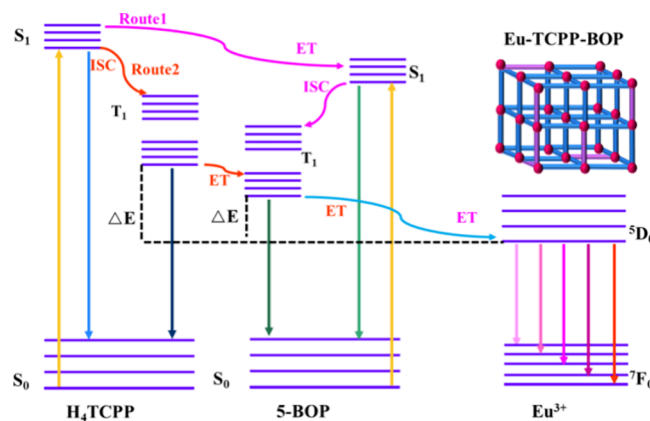


Figure 8. Schematic illustration of the absorption, migration, and emission from Eu-MOFs with the antenna effect procedure.

Analytical Performance of Eu-TCPP-BOP. Catechol (CC), a phenylene glycol derivative, is widely used in the production of plastics, rubber, pesticides, cosmetics, food additives, and pharmaceuticals.^{50–52} However, excessive catechol intake can cause health problems such as hypertension, kidney issues, respiratory irritation, and liver dysfunction.^{53,54} Moreover, catechol is resistant to dilution and decomposition, even at low concentrations, posing significant environmental risks and threats to both freshwater and marine ecosystems.^{55–57} Thus, the effective detection of catechol is crucial.

Figure 9a,b present the quantitative analysis of catechol concentrations (ranging from 0–70 μM) using Eu-TCPP-BOP as the sensing agent. In Figure 9b calibration curve demonstrates the relationship between the logarithm of ΔECL intensity and catechol concentration, described by the equation $\Delta\text{ECL} = 284.0 + 6614.4 \log(\text{C}/\mu\text{M})$, with an $R^2 = 0.9973$. The LOD was calculated to be 0.35 μM ($\text{S/N} = 3$) using the formula $\text{LOD} = 3\sigma/K$, where σ is the standard deviation of three measured signals, and K represents the slope of the standard curve. Compared to previously published studies (Table S1), this sensor offers a broader linear range and a lower detection limit.

The reproducibility of the sensor was evaluated by adding 70 μM catechol to a mixture of 0.1 M $\text{K}_2\text{S}_2\text{O}_8$ and 0.1 M PBS/KCl ($\text{pH} = 7.4$), resulting in an RSD of 2.55%, highlighting its excellent reproducibility (Figure 9c). Additionally, the selectivity of Eu-TCPP-BOP was assessed by introducing various interfering substances, including Ca^{2+} , Zn^{2+} , Fe^{3+} , ascorbic acid (AA), lactose, sorbitol, and glucose (Glu), into the same solution. The ECL intensity of the Eu-TCPP-BOP sensor remained stable, demonstrating strong selectivity against these potential interferences (Figure 9d).

Determination of CC in Yellow River Water. To evaluate the practical applicability of the sensor, catechol concentrations in water samples from the Yellow River were analyzed. Samples were collected from Jinniu Wharf in the Anning District of Lanzhou City, Gansu Province, and were spiked with standard catechol solutions at varying concentrations (15, 25, and 30 μM) for recovery analysis. Figure 9e presents the results of this recovery experiment. The average recovery rates for the Yellow River water samples ranged from 95.10% ~ 106.02%, with an RSD between 0.49% ~ 1.76%. These findings demonstrate that the Eu-TCPP-BOP sensor

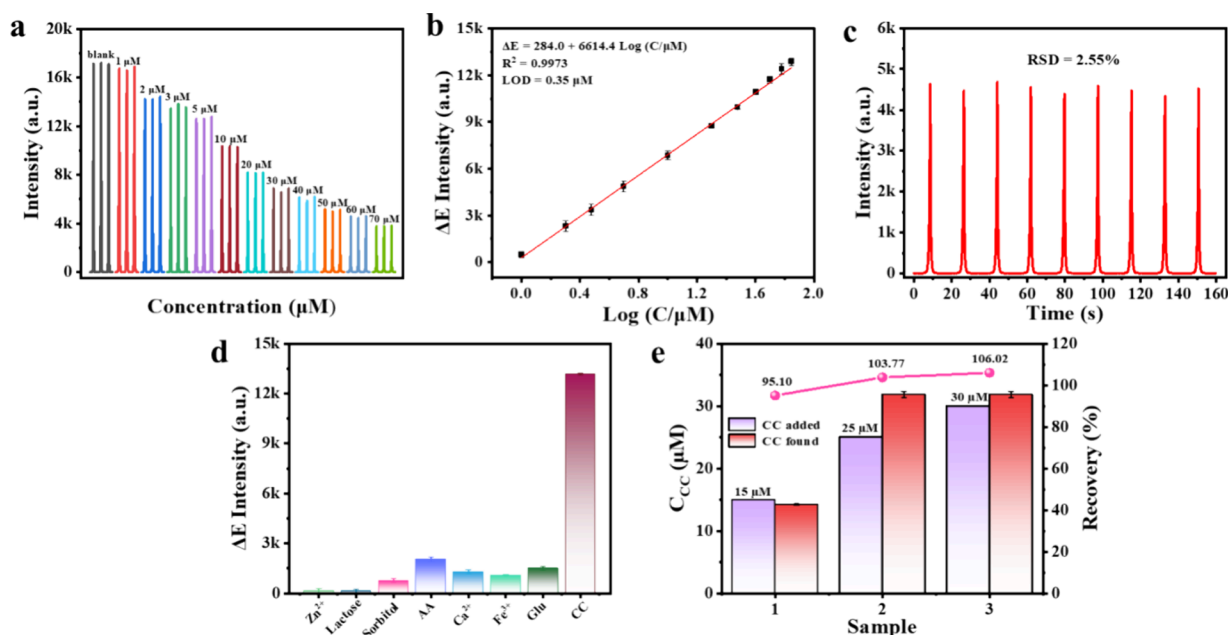


Figure 9. (a) ECL-time curves of Eu-TCPP-BOP/GCE for different concentrations of CC (0, 1, 2, 3, 16, 5, 10, 20, 30, 40, 50, 60, and 70 μM); (b) calibration plot of ΔECL intensity and logarithm of CC concentration; (c) ECL stability of Eu-TCPP-BOP/GCE scanning for nine cycles in the presence of 70 μM CC; (d) ECL response of Eu-TCPP-BOP/GCE to different substances (Ca^{2+} , Zn^{2+} , Fe^{3+} , AA, lactose, sorbitol, and Glu); (e) detection of CC in real samples. Experimental conditions: 0.1 M $\text{K}_2\text{S}_2\text{O}_8$, 0.1 M PBS/KCl (pH = 7.4), scanning rate 0.2 V/s, potential window 0.0 to -1.8 V (vs Ag/AgCl).

provides excellent reproducibility and practicality, making it well-suited for detecting catechol in real water samples.

CONCLUSIONS

In this study, we developed a mixed-ligand MOF, Eu-TCPP-BOP, which functions as a self-enhancing luminescent group for ECL. Scanning photoelectrochemical microscopy was employed to investigate the role of boron ligands as a bridge for electron transmission in enhancing the ECL performance of Eu-TCPP. The ligand 5-BOP acts as a mediator, facilitating electron transfer between the donor and the acceptor during the formation of Eu-TCPP-BOP, effectively channeling electron energy to the excited state of Ln^{3+} , thus optimizing energy transfer within the Ln-MOF. Under voltage excitation, Eu-TCPP-BOP exhibits efficient electron injection, producing a high and stable ECL signal. Using Eu-TCPP-BOP as the sensing platform, we developed a “signal off” ECL sensor for catechol detection. This approach offers a new analytical method for the sensitive detection of catechol and holds great promise for applications in water sample analysis.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.analchem.4c06857>.

FT-IR, XPS spectrum, UV-vis spectra, FL spectra, DPV curve, SEM image, XRD patterns, image and CV of UME, and performance comparison (PDF)

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Notes

The authors declare no competing financial interest.

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